

Experiment 1: Data Wrangling I

```
pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
from google.colab import files
import pandas as pd
```

```
uploaded = files.upload()
if uploaded:
    print("Success")
else:
    print("No file uploaded")
```

[Choose Files](#) StudentsPerformance.csv
StudentsPerformance.csv(text/csv) - 57019 bytes, last modified: 3/29/2026 - 100% done
 Saving StudentsPerformance.csv to StudentsPerformance.csv
 Success

```
import pandas as pd
```

```
df = pd.read_csv("/content/StudentsPerformance.csv")
```

```
print(df)
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	
..	...	...	...	...	
995	female	group E	master's degree	standard	
996	male	group C	high school	free/reduced	
997	female	group C	high school	free/reduced	
998	female	group D	some college	standard	
999	female	group D	some college	free/reduced	

	test preparation course	math score	reading score	writing score
0	none	72	72	NaN
1	completed	69	90	88.0
2	none	90	95	93.0
3	none	47	57	44.0
4	none	76	78	75.0
..	...	...	...	...
995	completed	88	99	95.0
996	none	62	55	55.0
997	completed	59	71	65.0
998	completed	68	78	77.0
999	none	77	86	86.0

```
[1000 rows x 8 columns]
```

find Missing Values

```
df.isnull()
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	False	False	False	False	False	False	False	True
1	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...
995	False	False	False	False	False	False	False	False
996	False	False	False	False	False	False	False	False
997	False	False	False	False	False	False	False	False
998	False	False	False	False	False	False	False	False
999	False	False	False	False	False	False	False	False

1000 rows × 8 columns

df.isnull().sum()

	0
gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	0
reading score	0
writing score	1

dtype: int64

- remove columns that contain missing (NaN) values from a Pandas DataFrame.

```
# axis=1
#Specifies columns.
#axis=0 → rows
#axis=1 → columns
```

df.dropna(axis=1)

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score
0	female	group B	bachelor's degree	standard	none	72	72
1	female	group C	some college	standard	completed	69	90
2	female	group B	master's degree	standard	none	90	95
3	male	group A	associate's degree	free/reduced	none	47	57
4	male	group C	some college	standard	none	76	78
...	...	...	...	...	...	...	...
995	female	group E	master's degree	standard	completed	88	99
996	male	group C	high school	free/reduced	none	62	55
997	female	group C	high school	free/reduced	completed	59	71
998	female	group D	some college	standard	completed	68	78
999	female	group D	some college	free/reduced	none	77	86

1000 rows × 7 columns

Describe Method to get Initial Statistics

```
print(df.describe())
```

	math score	reading score	writing score
count	1000.00000	1000.000000	999.000000
mean	66.08900	69.169000	68.048048
std	15.16308	14.600192	15.202102
min	0.00000	17.000000	10.000000
25%	57.00000	59.000000	57.500000
50%	66.00000	70.000000	69.000000
75%	77.00000	79.000000	79.000000
max	100.00000	100.000000	100.000000

Data Formating: Display Data types if not in proper format apply proper conversion method

```
df.dtypes
```

	0
gender	object
race/ethnicity	object
parental level of education	object
lunch	object
test preparation course	object
math score	int64
reading score	int64
writing score	float64

dtype: object

integer-location based indexing (select specific columns from a DataFrame using index positions)

```
y = df.iloc[:, 0:2]
print(y)
```

	gender	race/ethnicity
0	female	group B

```

1   female      group C
2   female      group B
3   male        group A
4   male        group C
..   ...
995 female      group E
996 male        group C
997 female      group C
998 female      group D
999 female      group D

```

[1000 rows x 2 columns]

Turn Categorical Variable into Numerical

```

from sklearn.preprocessing import LabelEncoder
y = ['male', 'female']
le = LabelEncoder()
y = le.fit_transform(y)
print(y)

```

[1 0]

#Count how many times each unique value appears in the column race/ethnicity of a Pandas DataFrame.

```
print(df['race/ethnicity'].value_counts())
```

```

race/ethnicity
group C    319
group D    262
group B    190
group E    140
group A     89
Name: count, dtype: int64

```

convert a categorical column (lunch) into numerical form using One-Hot Encoding and then add it back to the origir
pd.get_dummies() performs One-Hot Encoding.
It converts each unique category in the lunch column into a separate binary (0/1) column.

```

df_Lunch = pd.get_dummies(df['lunch'])
df_new = pd.concat([df, df_Lunch], axis=1)#pd.concat() joins DataFrames together.axis=1 means join column-wise.
#This adds the new one-hot encoded columns to the original DataFrame.
print(df_new)

```

```

gender race/ethnicity parental level of education      lunch \
0   female      group B      bachelor's degree      standard
1   female      group C      some college      standard
2   female      group B      master's degree      standard
3   male        group A      associate's degree  free/reduced
4   male        group C      some college      standard
..   ...
995 female      group E      master's degree      standard
996 male        group C      high school    free/reduced
997 female      group C      high school    free/reduced
998 female      group D      some college      standard
999 female      group D      some college    free/reduced

test preparation course  math score  reading score  writing score \
0                none          72          72          NaN
1            completed          69          90          88.0
2                none          90          95          93.0
3                none          47          57          44.0
4                none          76          78          75.0
..   ...
995            completed          88          99          95.0
996                none          62          55          55.0
997            completed          59          71          65.0
998            completed          68          78          77.0
999                none          77          86          86.0

free/reduced  standard
0          False      True
1          False      True
2          False      True
3           True     False
4          False      True

```

```
..      ...      ...
995      False      True
996      True      False
997      True      False
998      False      True
999      True      False
```

[1000 rows x 10 columns]

```
from sklearn.preprocessing import LabelEncoder

y = ['Yes', 'No', 'Yes', 'No'] # define y first

le = LabelEncoder()
y = le.fit_transform(y)

print(y)

[1 0 1 0]
```

#LabelEncoder converts labels (categories) into numeric values.

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("StudentsPerformance.csv")

y = df['gender'] # column name containing labels

le = LabelEncoder()
y = le.fit_transform(y) #fit(): finds unique classes in y transform(): assigns numbers to each class.

print(y)
```

```
[0 0 0 1 1 1 0 0 1 1 0 1 1 0 1 1 0 1 0 0 1 0 1 0 1 0 1 1 1 1 0 1 0 0 0 0 0 1 1 1 1 0
0 0 1 1 0 0 1 0 1 0 0 0 1 1 1 1 1 0 0 0 1 1 0 1 1 1 0 0 1 1 0 1 0 0 1 0 0 1 0 1
1 1 1 1 0 0 0 1 1 1 1 0 0 0 0 0 0 0 1 1 1 0 1 1 0 0 0 1 1 0 1 1 0 0 1 0 0 0 1 0 0 0
1 1 0 0 1 0 0 0 0 0 1 0 1 1 0 1 1 1 0 1 1 1 1 0 1 1 0 1 0 0 0 1 1 0 1 1
0 1 1 1 0 1 1 0 0 1 0 1 1 0 1 1 0 0 1 0 0 0 1 1 0 0 0 0 0 0 0 0 1 0 0 0 1
1 1 1 1 0 0 1 0 1 0 1 1 1 1 0 0 0 0 1 0 1 1 1 1 0 0 1 1 0 1 1 1 0 0 1 1 0 1
0 1 0 0 0 1 1 0 1 1 0 1 1 1 1 0 1 1 1 0 0 1 1 1 1 0 0 1 1 0 0 1 1 0 0 1 0
0 0 1 0 0 1 1 0 0 0 0 1 1 0 0 1 1 1 0 0 0 1 1 1 0 0 0 1 1 0 1 1 1 1 0 1 1
1 1 1 1 1 0 1 0 1 1 1 0 0 0 1 1 0 0 1 0 1 1 0 0 0 0 0 0 1 1 1 0 1 1 1
1 0 0 1 1 0 0 1 0 0 1 1 0 1 1 0 1 0 1 0 0 0 0 1 0 1 0 0 1 1 1 1 0 0
1 0 1 0 0 1 0 0 0 1 0 1 1 0 0 0 0 0 0 1 1 0 1 1 0 1 0 0 1 1 0 0 0 0 1
0 0 1 0 1 1 1 0 1 1 1 1 1 0 0 0 0 1 0 1 1 1 1 0 1 1 1 1 1 0 0 0 0
1 0 1 1 1 1 0 0 0 1 0 1 0 1 0 1 1 1 0 0 1 0 0 1 0 0 0 0 0 1 1 0 1 1
0 1 1 0 1 1 0 1 1 0 0 0 0 0 1 0 0 0 1 0 0 1 0 0 0 1 1 1 0 1 1 0 0 0 0
0 0 1 0 1 1 1 1 1 0 0 0 0 0 1 0 1 0 1 0 1 1 1 1 0 0 0 1 0 1 0 1 1 0 1 1
0 0 1 0 1 0 0 1 0 1 1 0 0 1 1 1 1 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 1 1
1 0 0 0 1 1 0 0 0 0 1 1 1 0 0 0 0 1 0 1 0 0 0 0 1 1 1 0 1 1 1 1 1
0 1 1 0 0 1 1 0 0 1 0 1 0 0 0 1 0 0 0 0 0 1 0 0 0 0 1 1 0 1 1 1 0 0 1 0
0 0 1 1 0 1 0 0 0 0 0 0 1 1 0 1 1 0 1 0 1 1 1 1 0 0 0 0 0 0 0 0 1 0 0 1
0 0 1 1 1 1 0 1 0 1 0 0 1 0 0 1 0 1 1 1 0 1 0 1 0 1 0 1 1 0 1 1
1 0 0 0 1 1 1 1 0 1 1 1 1 0 1 0 1 1 0 1 0 1 0 1 0 1 0 1 1 1 0 0 1 0 0
0 0 1 0 0 0 0 1 0 0 0 1 0 0 0 1 1 0 0 1 0 1 1 1 0 0 0 1 0 0 1 1 1 0 1
0 1 0 1 0 0 0 0 1 0 0 1 0 0 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 1 1 0 1 1
0 0 1 1 0 1 0 1 1 0 1 0 1 1 1 1 1 1 0 1 1 1 0 0 1 0 0 1 0 0 0 1
0 1 0 0 0 1 0 0 1 0 1 0 0 0 0 1 1 0 0 1 1 0 0 0 0 0 1 0 0 1 1 0 1 0 1
1 1 0 1 0 1 1 1 1 1 1 1 0 1 1 1 0 1 1 0 0 1 0 1 0 1 1 0 0 1 0 0
0 0 1 0 1 1 0 0 0 1 0 0 0 0 1 1 1 0 0 1 1 0 0 1 0 1 0 0 0 1 0 0 0
0 1 0 1 1 0 0 0 1 0 0 0 0 1 1 1 0 0 1 1 0 0 1 0 0 0 1 0 0 1 0 0 0]
```

df.head()

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score	
0	female	group B	bachelor's degree	standard	none	72	72	NaN	
1	female	group C	some college	standard	completed	69	90	88.0	
2	female	group B	master's degree	standard	none	90	95	93.0	